

# *Probability in Modern Scanning Systems: Barcodes and QR Codes*

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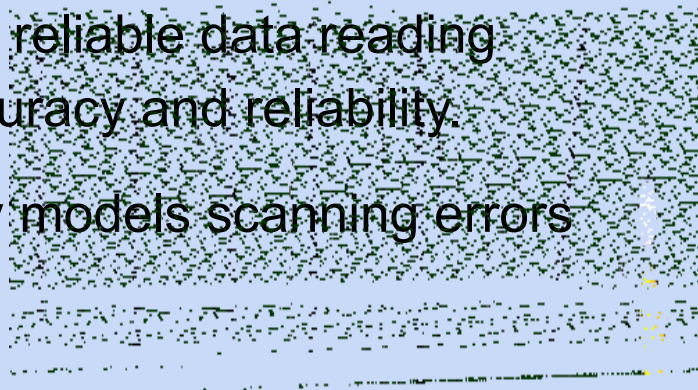
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# Introduction

- Barcodes and QR codes are essential for efficient data processing in automated systems, digital commerce, and logistics.
- They rely on **probabilistic models** to ensure reliable data reading under **uncertainty and noise**, improving accuracy and reliability.
- This project explores how **probability theory** models scanning errors and improves data reliability.



- **Problem Statement:**

- Modern systems such as supermarkets, online shopping, factories, and logistics handle large amounts of data.
- Manual data entry is slow and prone to errors, which may cause financial loss or system failures.
- **Probabilistic error models** are essential to prevent data loss in noisy environments.

- **Why Barcodes and QR Codes Matter:**

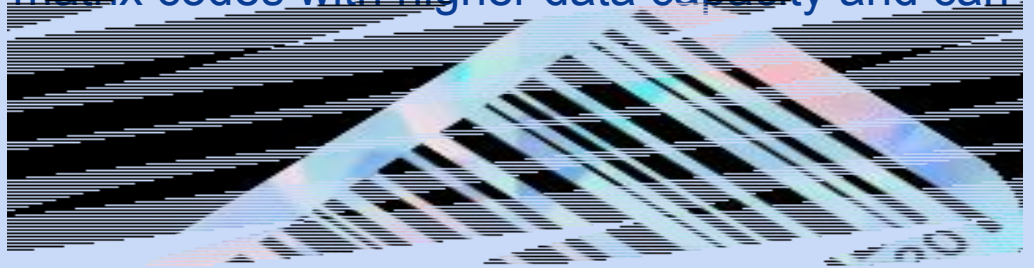
- Barcodes and QR codes encode information in a way that allows fast scanning, **error detection**, and **error correction**.
- Probability theory helps increase the **chance of correct decoding**, even when part of the code is damaged.

- **Motivation:**

- Studying barcodes and QR codes shows how **probability concepts** are used to model random scanning errors and improve system reliability in real-world applications.

# What Are Barcodes and QR Codes?

- Barcodes are one-dimensional representations of data using black and white stripes.
- QR codes are two-dimensional matrix codes with higher data capacity and can be read from any orientation.



# How Do They Work?

- Scanning may introduce **random errors** due to noise or distortion
- Error detection estimates whether the data is likely correct
- Error correction increases the **probability of successful decoding**
- The scanner selects the **most probable original data**

## Probability in Error Detection & Correction:

- Each bit has a **probability of error**  $p$  due to noise
- Redundancy increases **probability of correct decoding**.

# The Role of Probability

- **Probability in Scanning**

- Scanning errors occur randomly due to lighting, damage, or motion.
- Probability models the chance of bit errors.

- **Error Correction**

- Extra data is added to increase the probability of correct recovery.
- Even with missing parts, decoding can succeed.

- **System Reliability**

- Probability helps measure success rates of scanning systems.

- *Bit errors follow a **binomial distribution** in noisy channels.*

- $P(\text{success}) = 1 - P(\text{failure})$

$$p(0) = 1 - p$$

$$p(1) = p$$

# Code for creating a qr code

(This code demonstrates how encoding and redundancy increase the probability of successful data recovery during scanning.)

```
!pip install qrcode
import numpy as np
import matplotlib.pyplot as plt
import qrcode
from PIL import Image

# Encodes data into qr format
data = "Math 205 Project!"
qr = qrcode.QRCode(box_size=10)
qr.add_data(data)
qr.make(fit=True)

# Creates the qr code image
qr_code = qr.make_image(fill="black", back_color="white")
plt.imshow(qr_code, cmap='gray')
plt.axis("off")
plt.show()
```



# Code 128 Barcode Addresses

(This implementation includes **redundancy** to increase the **probability of successful recovery** in noisy scans.)

```
# Code 128 barcode standard does not share the method to derive their addresses algorithmically
```

```
# So it is impossible to translate characters into the correct addresses without using a hard coded list
```

```
code128_patterns = [  
    "11011001100", "11001101100", "11001100110", "10010011000", "10010001100", "10001001100",  
    "10011001000", "10011000100", "10001100100", "11001001000", "11000100100", "11000100100",  
    "10110011100", "10011011100", "10011001110", "10111001100", "10011101100", "10011100110",  
    "11001110010", "11001011100", "11001001110", "11011100100", "11001101000", "11101101110",  
    "11101001100", "11100101100", "11100100110", "11101100100", "11100110100", "11100110010",  
    "11011010000", "11011000110", "11000110110", "10100011000", "10001011000", "10001000110",  
    "10100010000", "10001101000", "10001100010", "11010001000", "11000101000", "11000100010",  
    "10110110000", "10110001110", "10001101110", "10111011000", "10111000110", "10001110110",  
    "11101110110", "11010001110", "11000101110", "11011101000", "11011100010", "11011101110",  
    "11101010000", "11101000110", "11100010110", "11101101000", "11101100010", "11100011010",  
    "11101111010", "11001000010", "11110001010", "10100110000", "10100001100", "10010110000",  
    "10010000110", "10000101100", "10000100110", "10110010000", "10110000100", "10011010000",  
    "10011000010", "10000110100", "10000110010", "11000010010", "11001010000", "11110111010",  
    "11000010100", "10001111010", "10100111100", "10010111100", "10010011110", "10111100100",  
    "10011110100", "10011110010", "11110100100", "11110010100", "11110010010", "11011011110",  
    "11011110110", "11110110110", "10101111000", "10100011110", "10001011110", "10111101000",  
    "10111100010", "11110101000", "11110100010", "10111011110", "10111101110", "11101011110",  
    "11110101110", "11010000100", "11010010000", "11010011100", "11000111011" ]
```

```
# Maps characters to the list
```

```
def char_to_128(char):
```

```
    ascii_val = ord(char)
```

```
    if 32 <= ascii_val <= 126:
```

```
        return ascii_val - 32
```

```
    else:
```

```
        raise ValueError(f"Unsupported character '{char}' for Code 128-B.")
```



# Encoding a Barcode

```
def encode_code128bar(text):  
    start_code = 104 # Start code  
    codes = [start_code] + [char_to_128(c) for c in text]  
  
    # The checksum is calculated by the formula (Start + char1 * 1 + char2 * 2 + ...) % 103  
    checksum = codes[0]  
    for i, code in enumerate(codes[1:], start=1):  
        checksum += code * i  
    checksum %= 103  
  
    codes.append(checksum)  
    codes.append(106) # Stop code  
  
    # Convert to barcode pattern  
    pattern = ''.join(code128_patterns[code] for code in codes)  
    print(pattern)  
    return pattern
```

# Creating a Barcode Image

(The checksum and encoding process reduce the probability of decoding errors.)

```
def render_barcode(binary_pattern, height=60, bar_width=3):  
    width = len(binary_pattern) * bar_width  
    image = np.ones((height, width), dtype=np.uint8) * 255  
  
    for i, bit in enumerate(binary_pattern):  
        if bit == '1':  
            image[:, i * bar_width:(i + 1) * bar_width] = 0  
  
    return Image.fromarray(image)
```

# Example

```
text = "Math 205 Project!"  
pattern = encode_code128bar(text)  
barcode = render_barcode(pattern)  
plt.imshow(barcode, cmap='gray')  
plt.title("QR Code Binary Matrix")  
plt.axis("off")  
plt.show()
```



# Applications of “Barcodes”

- **Retail systems:** reduce probability of scanning errors and increase checkout accuracy
- **Logistics:** improve delivery accuracy and reduce probability of misrouting
- **Damaged codes:** error correction allows decoding even with partial damage
- **Payments & tickets:** increase reliability and reduce transaction failures
- **Tracking systems:** lower the probability of data loss during scanning

## Limitations & Considerations:

- **Poor lighting** reduces scan reliability
- **Severe damage** can prevent correct decoding
- **Large data** requires larger QR codes
- **Stronger error** correction improves reliability but reduces data capacity
- **Probabilistic trade-off:** stronger error correction reduces data capacity but increases reliability.

# Probability Models in QR Codes

## Random Errors in Scanning

- Each scanned bit has a probability  $p$  of being incorrect due to noise, lighting, or damage.
- Bit errors can be modeled as **independent Bernoulli trials**.

## Error Probability for Multiple Bits

- For a message of  $n$  bits, the probability that **at least one bit is in error** is:  $P(\text{error in message}) = 1 - (1 - p)^n$
- Where:
  - $p$  = probability of a single bit error
  - $n$  = total number of bits in the message
  - $(1 - p)^n$  = probability that all  $n$  bits are correct

## Redundancy and Reliability

- By adding redundant bits (error correction), the system can **recover the original data** even if some bits are corrupted.
- The **expected reliability** of a QR code scanning system is:  $P(\text{successful decode}) = 1 - P(\text{uncorrectable error})$
- Higher redundancy  $\rightarrow$  lower  $P(\text{uncorrectable error}) \rightarrow$  higher reliability.
- Example:
  - If  $p = 0.01$  (1% chance of bit error) and  $n = 100$ :  $P(\text{error in message}) = 1 - (0.99)^{100} \approx 0.634$
  - Without correction, there's a **63.4% chance** the message contains at least one error.
  - With error correction, this probability drops significantly.

- **Error Correction (Probabilistic View)**
  - Some parts of the code may be damaged
  - Error correction increases the probability of recovering original data
  - The more redundancy added, the higher the success rate

# Reed–Solomon (Simple Probability View)

- Data is transmitted with additional redundant symbols
- During scanning, some symbols may be lost or corrupted randomly
- Error correction uses redundancy to **reduce the probability of decoding failure**
- If the number of errors is small, the original data can still be recovered
- This greatly increases the **reliability** of QR code scanning
- If  $k$  errors occur, the **probability of recovery** depends on redundancy level  $r$ , where  $r \geq 2k$ .

# Example: Reed-Solomon in Action

- Assume a message:

[32, 91, 11, 120, 209]

- Using Reed-Solomon, we generate extra symbols: [57, 12]

- These redundant symbols allow us to correct **1 error** in the original message.

- Even if a symbol is damaged or missing, the full message can be **reconstructed**.

[32, 91, 11, 120, 209]

+ Add → [57, 12]

→ Final Code Sent

→ Error Happens

→ Error Corrected → [Original Message Recovered]



# Conclusion

- Barcodes and QR codes rely on probability theory to handle random errors during scanning. By modeling uncertainty and increasing reliability, these systems ensure accurate data transmission in real-world conditions.
- This project demonstrates how **probability theory** enables reliable data transmission under uncertainty.



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